

Hotel Booking and Cancellation Behaviour Analysis

1. Introduction

- In this project, I analysed hotel booking data to understand customer booking behaviour and cancellation patterns.
- The dataset contains hotel booking information for **City Hotel** and **Resort Hotel**. The main purpose of this analysis was to understand which hotel type customers prefer and which booking factors are related to a higher cancellation rate.

The project focuses on the following main business questions:

1. Which hotel type receives more bookings?
2. How do booking trends change across different months?
3. Which hotel type has a higher cancellation risk?
4. Does the length of stay affect the cancellation rate?
5. Does lead time affect the cancellation rate?
6. Which combination of booking characteristics may indicate a higher risk of cancellation?

Answering these questions can help hotel management understand customer behaviour, improve demand planning, reduce cancellations, and make better operational decisions.

Before performing detailed analysis, key performance indicators were calculated to provide a high-level overview of the hotel booking dataset.

The main KPIs analysed were:

- Total valid bookings after data cleaning.
- Overall cancellation rate.
- Average booking lead time.
- Average Daily Rate (ADR).

These indicators provide an overall understanding of hotel booking behaviour and create a baseline for the detailed analysis presented in the following sections.

The remaining analysis investigates how booking patterns, hotel type, stay duration, and lead time are related to customer cancellation behaviour.

2. Hotel Booking Analysis

Which Hotel Type Is Booked More Often?

- The analysis shows that **City Hotel is booked more often than Resort Hotel**.
- City Hotel represents approximately **60.98% of total bookings**, while Resort Hotel represents approximately **39.02%**.
- This means that City Hotel received a larger share of customer bookings in this dataset.
- One possible reason is that City Hotels can be useful for different types of travel, such as business trips, short visits, events, and tourism. Resort Hotels may depend more on leisure travel and vacation periods.

Monthly Booking Pattern

- I also analysed how the number of bookings changes during different months.
- The booking numbers show that hotel demand changes throughout the year. Both City Hotel and Resort Hotel have months with higher and lower booking demand.
- For both hotel types, **October had the highest number of bookings**.
- **City Hotel**
- The highest number of bookings was recorded in **October, with 6,488 bookings**.
- The lowest number of bookings was recorded in **March, with 2,692 bookings**.

Resort Hotel

The highest number of bookings was also recorded in **October, with 4,647 bookings**.

The lowest number of bookings was recorded in **March, with 1,932 bookings**.

This shows that hotel bookings are affected by seasonal patterns.

Possible reasons for this pattern include:

- Holidays and vacation periods
- Changes in tourism demand
- Weather conditions
- Festivals and special events
- Business travel

Business Recommendation Based on Monthly Demand

Hotels can use these booking patterns to plan their operations.

During high-demand months, hotels can:

- Prepare enough staff.
- Manage room availability carefully.
- Increase prices when demand is high.
- Prepare for a larger number of guests.

During low-demand months, hotels can:

- Offer discounts or promotional packages.
- Run marketing campaigns.
- Schedule maintenance work.
- Create offers to attract more customers.

Note: The monthly booking analysis combines bookings from all available years in the dataset. Therefore, the results represent general seasonal booking patterns rather than a separate year-by-year booking trend.

3. Stay Duration and Cancellation Behaviour

- The next part of the analysis examined whether the total number of nights stayed by a customer is related to the cancellation rate.
- I calculated the total stay duration using both weekend nights and weekday nights.
- The cancellation rates were then compared for City Hotel and Resort Hotel.

City Hotel Results

For City Hotel, the cancellation rate generally increased as the stay duration increased.

Stay Duration	Cancellation Rate
1 Night	21.98%
2–3 Nights	30.70%
4–5 Nights	31.64%
6–7 Nights	37.53%
More than 7 Nights	55.03%

- The highest cancellation rate was found for bookings with a stay of more than seven nights.
- This shows a **positive relationship** between stay duration and cancellation rate for City Hotel. In simple words, longer planned stays generally had a higher chance of being cancelled.

Resort Hotel Results

For Resort Hotel, cancellation rates were:

Stay Duration	Cancellation Rate
1 Night	12.05%
2–3 Nights	23.00%
4–5 Nights	28.34%
6–7 Nights	29.03%

Stay Duration	Cancellation Rate
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More than 7 Nights	27.63%
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- The cancellation rate generally increased as stay duration increased, although the pattern was not completely consistent for stays longer than seven nights.

Comparison Between the Two Hotel Types

- The relationship between stay duration and cancellation was stronger for City Hotel.
- City Hotel showed a large increase in cancellation rates for longer stays. Resort Hotel also showed an increase, but its cancellation rates remained lower than City Hotel.

Possible Reasons

- There are several possible reasons why longer stays may have a higher cancellation rate.
- **First**, a longer stay usually costs more money. Customers may change their decision if the total cost becomes too high.
- **Second**, longer trips usually require more planning. When customers plan a trip far in advance, their personal or work plans may change before the arrival date.
- **Third**, customers may book a hotel before their travel plan is fully confirmed and later cancel the booking.

Recommendation

- Hotels should pay more attention to long-stay bookings, especially for City Hotel.
- Hotels can send confirmation messages or reminders to customers before their arrival date. They can also consider appropriate cancellation terms for long stays.

4. Lead Time and Cancellation Behaviour

- Lead time means the number of days between the date when a customer makes a booking and the planned arrival date.
- For example, if a customer books a hotel 100 days before arrival, the lead time is 100 days.
- I analysed whether bookings made earlier have a higher or lower cancellation rate.

City Hotel Results

Lead Time	Cancellation Rate
0–7 Days	10.54%
8–30 Days	27.96%
31–90 Days	33.42%
91–180 Days	36.73%
181–365 Days	45.18%
More than 365 Days	52.82%

- The results show a clear pattern.
- For City Hotel, cancellation rates increased as lead time increased.
- Bookings made within 7 days of arrival had a cancellation rate of only **10.54%**. However, bookings made more than 365 days before arrival had a cancellation rate of **52.82%**.
- This means that City Hotel bookings made far in advance had a much higher risk of cancellation.

Resort Hotel Results

Lead Time	Cancellation Rate
0–7 Days	5.95%
8–30 Days	20.83%
31–90 Days	29.68%

Lead Time	Cancellation Rate
91–180 Days	32.13%
181–365 Days	34.84%
More than 365 Days	21.39%

- For Resort Hotel, cancellation rates also generally increased as lead time increased.
- However, the cancellation rate decreased for bookings made more than 365 days in advance.
- This result should be interpreted carefully because this group had only **173 bookings**, which is a much smaller number compared with the other groups.

Comparison Between the Two Hotels

- Both hotels showed the same general pattern: bookings made closer to the arrival date had lower cancellation rates, while bookings made further in advance generally had higher cancellation rates.
- However, City Hotel showed a much stronger increase in cancellation rate.
- City Hotel increased from **10.54%** for bookings made within 7 days to **52.82%** for bookings made more than 365 days in advance.
- Resort Hotel also increased as lead time increased, but the change was less extreme.

5. Combined Cancellation Risk Analysis

- The previous analyses examined stay duration and lead time separately. To gain a deeper understanding of cancellation behaviour, both variables were analysed together.
- This combined analysis helps identify booking profiles that may have a higher risk of cancellation.
- The analysis shows how cancellation rates change when different lead time groups are combined with different stay duration groups. This provides a more detailed view than analysing each factor separately.

High-Risk Booking Profile

- The combined analysis can help identify combinations of booking characteristics associated with higher cancellation risk.
- Bookings with longer lead times and longer planned stays may require additional attention because there is more time for customer travel plans to change before arrival.
- However, the exact high-risk booking groups should be identified based on the results shown in the combined cancellation risk heatmap.

Business Impact

Hotels can use this analysis to identify bookings that may require additional confirmation or follow-up.

Possible actions include:

- Sending booking confirmation reminders.
- Asking customers to reconfirm their plans closer to the arrival date.
- Providing suitable rescheduling options.
- Applying appropriate deposit or cancellation policies for higher-risk booking profiles.

This approach can help hotels improve occupancy planning and reduce the financial impact of cancelled bookings.

6. Key Findings

After analysing the data, I found the following important points:

Finding 1: City Hotel Is More Popular

- City Hotel received approximately **61% of total bookings**, compared with approximately **39% for Resort Hotel**.

Finding 2: Booking Demand Changes During the Year

- The number of bookings changed across different months. October was the busiest month for both hotel types.
- This shows that hotels should consider seasonal demand when planning their operations.

Finding 3: Longer Stays Have Higher Cancellation Risk

- For City Hotel especially, longer stays were associated with higher cancellation rates.

Finding 4: Lead Time Has a Strong Relationship With Cancellation

- Bookings made far in advance generally had a higher cancellation rate.
- This was one of the clearest patterns found in the analysis.

Finding 5: Combined Booking Characteristics Can Help Identify Cancellation Risk

- Analysing lead time and stay duration together provides a more detailed understanding of cancellation behaviour.
- Instead of considering only one factor, hotel management can identify booking profiles based on a combination of characteristics.
- This can help hotels focus their cancellation management strategies on bookings that show relatively higher cancellation risk.

7. Business Recommendations

Recommendation 1: Focus More on Long Lead-Time Bookings

The strongest finding in this project is the relationship between lead time and cancellation.

Hotels should pay more attention to customers who book many months before their arrival date.

Possible actions include:

- Sending booking reminders.
- Asking customers to reconfirm their booking.
- Providing easy rescheduling options.
- Using suitable deposit or cancellation policies.

These actions may help reduce cancellations.

Recommendation 2: Create Different Strategies for City Hotel

- City Hotel had higher cancellation rates than Resort Hotel in the stay duration and lead time analysis.
- Therefore, City Hotel may need stronger cancellation management.
- The hotel can identify bookings with long lead times or long stays and send additional confirmation messages to these customers.

Recommendation 3: Use Monthly Trends for Planning

Hotels should use monthly booking patterns for:

- Staff planning
- Room availability planning
- Pricing decisions
- Marketing campaigns

During high-demand months, hotels can focus on increasing revenue.

During low-demand months, hotels can use discounts and promotional offers to attract customers.

Recommendation 4: Monitor Long-Stay Bookings

- Longer stays, especially at City Hotel, showed higher cancellation rates.
- Hotels should monitor these bookings and communicate with customers before the arrival date to confirm their travel plans.

Recommendation 5: Use Combined Booking Risk Indicators

Hotels should not rely on only one booking characteristic when monitoring cancellation risk.

Lead time and stay duration can be analysed together to identify booking profiles that may require additional attention.

For example, bookings that fall into higher-risk combinations can be prioritised for:

- Confirmation reminders.
- Reconfirmation requests.
- Personalised communication.
- Suitable cancellation or deposit policies.

Using multiple booking characteristics together can help hotels develop a more targeted approach to cancellation management.

8. Recommendation With the Biggest Expected Impact

- The recommendation that I believe could have the biggest impact is:
- **Improve cancellation management for bookings made far in advance.**
- This recommendation is supported by the data.
- For City Hotel, the cancellation rate increased from **10.54% for bookings made within 7 days** to **52.82% for bookings made more than 365 days in advance**.
- This is a very large difference.
- Therefore, hotels can use lead time as an important indicator to identify bookings that may have a higher risk of cancellation.
- By sending reminders, asking for reconfirmation, and providing suitable rescheduling or cancellation options, hotels may be able to reduce the number of cancelled bookings and improve occupancy planning.

9 . Final Conclusion

- This project analysed hotel booking behaviour and cancellation patterns using Python, data cleaning, exploratory data analysis, and data visualisation.
- The analysis showed that City Hotel received more bookings than Resort Hotel. Hotel demand also changed throughout the year, showing the importance of understanding seasonal booking patterns for operational planning.
- The cancellation analysis showed that stay duration and booking lead time were both related to cancellation behaviour. Longer stays, particularly for City Hotel, generally showed higher cancellation rates. Bookings made far in advance also showed a higher cancellation risk, making lead time one of the strongest indicators identified in the analysis.
- In addition, combining lead time and stay duration provides a more detailed way to identify booking profiles that may have a relatively higher risk of cancellation.
- Overall, this analysis demonstrates how hotel booking data can be transformed into useful business insights. The findings can help hotel management improve cancellation management, occupancy planning, staffing decisions, pricing strategies, room availability planning, and customer communication.